

Prediction of Maternal and Child Health Outcomes from the Bangladesh Demographic and Health Survey

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Abstract—Bangladesh has made substantial progress against maternal and child mortality over the past three decades, but the remaining risk is still considerable and unevenly distributed. Neonatal death occurs in 4% of live births, 28.7% of measured young children are stunted, and only 57.2% of recent births occur in a health facility. Those most exposed to these outcomes are often the least able to reach care promptly. At an initial community-health visit, the worker usually has no laboratory measurements; the available information is closer to the household and reproductive history already captured by a national survey. We examine how useful that information is for early risk screening. The 2022 and 2017–18 Bangladesh Demographic and Health Surveys are combined into a leakage-audited dataset of 121,067 records, and five outcomes covering delivery, the neonatal period and early childhood are predicted. We compare Logistic Regression, Random Forest, XGBoost, LightGBM, CatBoost, RealMLP and TabM, together with a greedy post-hoc ensemble and the pretrained transformer TabPFN-3. All splits are grouped by maternal identity. Test ROC-AUC is between 0.688 for underweight and 0.845 for facility delivery; the methods differ by no more than 0.07 on any target. Two checks temper this result. A model trained on 2017–18 preserves much of its ranking performance on 2022 but overpredicts prevalence by as much as 5.9 times, while sensitivity at a 30% flag rate differs 5.7-fold across wealth quintiles. The limiting factor therefore seems to be the information recorded in the survey rather than the choice of model.

Index Terms—maternal and child health, Demographic and Health Survey, clinical prediction, data leakage, temporal validation, algorithmic fairness, tabular learning, CatBoost, TabM, ensemble selection

I. INTRODUCTION

Bangladesh has substantially reduced maternal and under-five mortality, but the remaining burden is not shared evenly. Across the two survey rounds used here, neonatal death occurs in 4.0% of live births, 28.7% of measured children are stunted, and 57.2% of recent births take place in a health facility. Facility delivery ranges from 34.0% in the poorest wealth quintile to 82.6% in the richest.

During an initial antenatal contact, a community health worker can observe much the same broad characteristics represented in a household survey: age, parity, birth spacing,

education, wealth, water and sanitation, and previous reproductive history. Laboratory measurements and biomarkers are not assumed to be available. The question is whether these variables can provide a useful early ranking of risk and retain that usefulness across survey rounds and socio-economic groups.

Many Bangladesh DHS prediction studies use complete cases from a single module, focus on one outcome, and rely on random splits without formally checking for leakage. Near-perfect discrimination is hard to reconcile with the pooled accuracy reported in a recent DHS meta-analysis [1]. Here we use one harmonised pipeline for two rounds, split records by mother, select targets according to observed learnability, test transport over time, and examine performance by wealth stratum.

The study makes four contributions: (i) a pooled corpus checked against round-specific indicators; (ii) a five-part leakage taxonomy with measurements before and after each control; (iii) a comparison of seven current tabular learners plus a post-hoc ensemble; and (iv) operational and equity analyses at clearly defined screening thresholds.

II. RELATED WORK

Eight studies provide the context: six use Bangladesh DHS data directly, one combines six East African countries and several survey rounds, and one is a meta-analysis of expected performance. Seven appeared in 2024 or later.

A. Prediction Studies on Bangladesh DHS Data

Salam *et al.* [2] used ten algorithms to predict four or more antenatal visits for 5,128 BDHS 2022 women. A random-forest variant led with 69.5% accuracy and AUC 0.734, providing the closest external comparison for our facility-delivery target.

Noor *et al.* [3] predicted the maternal continuum of four antenatal visits, skilled facility delivery and postnatal contact within 48 hours. In the same 5,128 women, random forest reached 82% accuracy and AUC 0.889, while only 25.3% completed the continuum. We instead model stages separately, allowing the likely point of drop-out to be identified.

Zoha *et al.* [4] predicted caesarean section from 2,490 BDHS 2022 records and reported AUC 0.871 with 45.6% prevalence, well above the 10–15% considered clinically indicated by the World Health Organization. Because this signal may reflect health-system behaviour rather than obstetric need, we fitted but excluded the outcome from the final panel (Section III-B).

Sultana *et al.* [5] compared six algorithms on 3,192 complete BDHS 2022 mother–child records. XGBoost led at 80% accuracy and AUC 0.761, with SHAP highlighting gestational age and administrative division. The study also illustrates the attrition caused by requiring measured birth weight.

Naznin *et al.* [6] reported 98.75% accuracy and 99.79% AUROC for under-five mortality among 4,913 BDHS 2022 respondents. Such discrimination is unusual for socio-demographic predictors and can indicate pre-split resampling or post-outcome features. We therefore treat it as a cautionary comparison; it also motivated our leakage audit, where neonatal death falls from ROC-AUC 0.967 to 0.773 after removing an arithmetic identity.

Islam *et al.* [7] modelled stunting, wasting and underweight in BDHS 2017–18. XGBoost reached 81.7% accuracy and AUC 0.802 for stunting, but 76.2% accuracy and AUC 0.622 for wasting. This contrast is why we use PR-AUC rather than headline accuracy for model selection.

B. Pooling Rounds, and the Performance Envelope

Tadese *et al.* [8] pooled 27,691 children from DHS rounds between 2016 and 2021 across six East African countries to predict incomplete immunisation. XGBoost led at 79.0% accuracy and AUC 0.86. The multi-round design and mostly maternal/reproductive predictors are directly relevant here.

A meta-analysis of eleven DHS malnutrition studies [1] reports pooled accuracies of 68.92% for stunting, 84.39% for wasting and 73.60% for underweight, with substantial heterogeneity. These values set a realistic reference for survey-based prediction and show that a connected maternal-and-child panel remains relatively unexplored.

C. Tabular Modelling Practice

Gradient-boosted trees remain strong general-purpose baselines [9]–[11]. TabArena [12] benchmarks sixteen models on fifty-one datasets, ranks CatBoost first among individual learners, and finds that diverse post-hoc ensembles can improve on single models. This motivates our greedy ensemble (Section IV-D), following Caruana *et al.* [13]. RealMLP and TabM are newer neural tabular models [14], [15]; TabPFN [16] is a transformer pretrained on synthetic tabular tasks. TRIPOD+AI [17] and equality-of-opportunity criteria [18] motivate our external and subgroup checks.

D. Identified Gap

The reviewed studies leave five gaps: connected maternal-and-child prediction, round pooling under one pipeline, leakage auditing, treatment of clustering, and wealth-stratified performance. We address all five and treat the resulting lower but defensible performance as a finding rather than a failure.

TABLE I
ANALYSIS COHORTS AND TARGETS

Cohort	Targets	Records
births	neonatal death	112,546
mother	equity analysis	45,458
child	stunting, severe stunting, underweight	11,954
recent birth	facility delivery	10,773

III. DATASET AND EXPLORATORY ANALYSIS

A. Source and Nested Cohorts

We use BDHS 2022 (DHS-8) and BDHS 2017–18 (DHS-7) [19]. Their Individual and Birth recodes were merged through one pipeline, producing 121,067 records, 899 variables, 1,346 clusters and 45,458 mothers. Since DHS modules are not observed for every record, analyses use nested cohorts: broad mother variables, narrower birth histories, recent-birth maternity variables, and anthropometry for physically measured children.

Figure 1 shows outcomes ranging from common pathway measures to rare adverse events, so accuracy alone is misleading. Missingness is treated as survey structure within each cohort, not as an automatic reason to remove a column. Ten published indicators were recomputed by round; all were within four percentage points of the published values. Stunting, for example, was 23.22% versus 23.6% in 2022 and 30.72% versus 31.0% in 2017–18.

B. Targets and Selection

The final panel contains facility delivery, neonatal death, stunting, underweight and severe stunting, with positive rates of 57.2%, 4.0%, 28.7%, 22.4% and 7.9%. Of fifteen candidates, fever, wasting and diarrhoea were difficult to learn; caesarean section reflected clinical decision-making; and pregnancy loss depended on recall timing. Incomplete immunisation fell from ROC-AUC 0.944 to 0.59 after restricting to the 12–23 month denominator and removing age, exposing age leakage.

IV. METHODOLOGY

A. Cleaning and Leakage

DHS files combine valid measurements, meaningful zeros and nonresponse codes. We decode them into missing values, true zeros and common numeric units, and namespace cluster and maternal identifiers by round. Feature blocks cover demographic, socio-economic, empowerment, household, reproductive, healthcare-access and child variables; 96–118 features remain per target.

Five leakage forms are audited: post-outcome variables, same-moment co-measurements, definitional variables, arithmetic identities, and informative missingness. For neonatal death, retaining the fertility identity gives ROC-AUC 0.967, versus 0.773 after blocking it; a survivor-only age variable gives 0.987. Both values and observation patterns therefore require inspection.

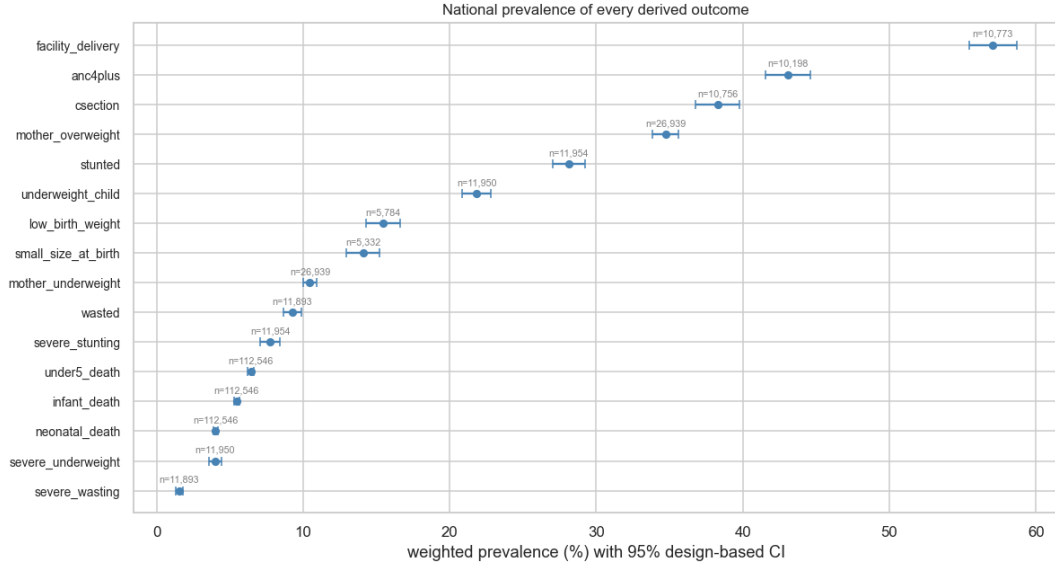


Fig. 1. Outcome prevalence with 95% design-based intervals. The broad range of base rates motivates target-specific metrics and thresholds.

TABLE II
SENSITIVITY TO LEAKAGE CONTROLS FOR NEONATAL DEATH

Feature set	ROC-AUC	Audit interpretation
Leakage-audited	0.773	Intended early-risk estimate
Fertility identity retained	0.967	Arithmetic label proxy
Survivor-only age retained	0.987	Missingness is the label

B. Feature Representation

Boosted trees use missingness natively. Linear and neural paths use median imputation with indicators, robust scaling and rare-level pooling; ordered DHS scales are ordinal-encoded. Splits are grouped by maternal identity because 90% of birth rows belong to mothers appearing more than once.

C. Sampling and Outcome Construction

Each target is constructed only within its valid DHS universe: facility delivery among births in the preceding five years, neonatal death from birth history, and anthropometric outcomes among children with DHS height and weight measurements. An unadministered module is not treated as a negative response. Feature selection follows the same cohort-specific rule.

Sampling weights estimate national prevalence and design-based intervals; cluster identifiers are retained for uncertainty checks. Held-out records are used for ROC-AUC and PR-AUC, so weights do not change groups or reveal outcomes during training. The average design effect is 2.19, making independent-row intervals about 1.48 times too narrow. Round-specific reconstruction guards against a pooled prevalence representing neither survey population.

TABLE III
COMPOSITION OF THE WINNING ENSEMBLES

Target	RF	XGB	LGBM	CatB	RealMLP	TabM
facility delivery	1	2	1	5	6	—
neonatal death	—	—	8	—	6	1
severe stunting	4	—	2	7	2	—

Entries are selection counts out of fifteen greedy rounds, so each row sums to fifteen; a dash means the learner was never selected and carries weight zero. Stunting and underweight are absent because a single model won those targets outright.

D. Models and the Ensemble

Seven learners are benchmarked: logistic regression; random forest; XGBoost; LightGBM; CatBoost [9]–[11]; RealMLP; and TabM [14], [15]. TabPFN-3 [16] is evaluated separately with the same split, preprocessing and row cap. All learners use published defaults; there is no hyperparameter search, so the comparison is between model families.

What “Ensemble” means here. This term denotes a defined blend. After fitting the seven learners, greedy forward selection with replacement [13] runs for fifteen rounds, appending the already-fitted learner that most improves PR-AUC on an inner validation split. A learner selected k times receives weight $k/15$; no new model or test-set weight is fitted. Table III reports the target-specific compositions.

Operating thresholds minimise an explicit false-negative cost while respecting a maximum flag rate of 30%, used here as a screening-capacity constraint.

E. Validation and Extensions

Internal evaluation uses grouped train–test splits and 400 bootstrap resamples for confidence intervals; temporal evaluation trains on 2017–18 and tests on 2022. Conformal prediction

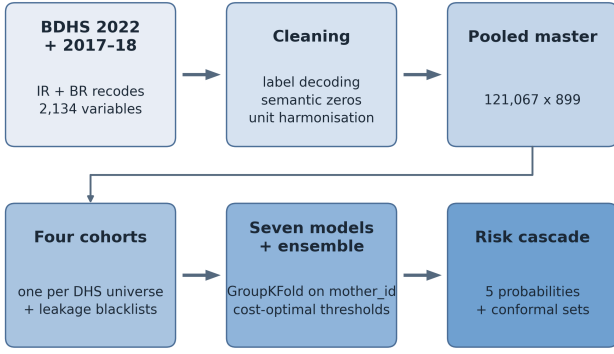


Fig. 2. Analysis pipeline. Survey recodes are decoded once, then passed through cohort-specific feature and leakage rules before grouped model fitting and risk reporting.

TABLE IV

BEST MODEL PER TARGET, WITH 95% BOOTSTRAP INTERVALS. **Ensemble** IS THE GREEDY PR-AUC-WEIGHTED BLEND OF THE SEVEN BENCHMARKED LEARNERS DEFINED IN SECTION IV-D

Target	Model	ROC-AUC	PR-AUC
facility delivery	Ensemble	0.845 [0.828, 0.859]	0.878
neonatal death	Ensemble	0.773 [0.756, 0.789]	0.209
stunted	CatBoost	0.707 [0.687, 0.730]	0.526
severe stunting	Ensemble	0.700 [0.660, 0.738]	0.192
underweight	TabM	0.688 [0.661, 0.712]	0.415

Winners are selected on PR-AUC. The composition of each ensemble is given in Table III; stunting and underweight were won by single models.

uses a held-out calibration split, survival analysis uses person-period rows, and multi-task learning jointly predicts the three nutrition outcomes. These are secondary analyses.

Figure 2 outlines the analysis from survey recodes to cohort-specific risk scores.

V. RESULTS

A. Internal Performance

Table IV lists the strongest model per target. ROC-AUC ranges from 0.688 to 0.845, with PR-AUC interpreted against each base rate. The full matrix in Table V shows close model families: the eight approaches span 0.025 ROC-AUC for facility delivery, 0.030 for stunting and 0.070 for neonatal death. In the last case, random forest’s recall falls to 0.001 at the default cut-point on a 4% base rate, indicating a threshold issue rather than failed ranking. The earlier nine-model panel changed no target by more than 0.02 ROC-AUC.

Two winners differ from the ROC-AUC leaders because selection uses PR-AUC. RealMLP leads stunting on ROC-AUC by 0.002, while CatBoost leads PR-AUC by 0.005; the ensemble leads underweight on ROC-AUC by 0.003, while TabM leads PR-AUC by 0.003. At a 22.4% base rate, PR-AUC is more relevant to screening.

At cost-sensitive operating points, recall increases from 0.167 to 0.502 for underweight and from 0.123 to 0.425

TABLE V

TEST ROC-AUC BY LEARNER AND TARGET. **Ensemble** IS NOT AN EIGHTH LEARNER: IT IS THE GREEDY PR-AUC-WEIGHTED BLEND OF THE SEVEN ROWS ABOVE IT

Learner	Facility delivery	Neonatal death	Stunted	Severe stunting	Underweight
Logistic Reg.	0.839	0.719	0.686	0.687	0.681
Random Forest	0.822	0.703	0.683	0.679	0.680
XGBoost	0.820	0.742	0.678	0.671	0.636
LightGBM	0.840	0.771	0.703	0.665	0.672
CatBoost	0.843	0.748	0.706	0.698	0.679
RealMLP	0.841	0.750	0.708	0.696	0.687
TabM	0.842	0.707	0.702	0.675	0.687
Ensemble	0.845	0.773	0.705	0.700	0.690
range	0.025	0.070	0.030	0.035	0.054

The ensemble is defined in Section IV-D and composed in Table III. Best per column in bold; the last row is the spread between the largest and smallest value above it. Values are point estimates on the held-out split, whereas Table IV reports bootstrap medians, so the two can differ in the third decimal.

for severe stunting, with more false alarms. Split conformal prediction provides 89.3–90.9% coverage at nominal 90%.

B. A Pretrained Transformer Changes Little

TabPFN-3 was tested with the same grouped split, native-missingness preprocessing and 10,000-row context cap. Its PR-AUC values were 0.872, 0.166, 0.516, 0.185 and 0.417 for facility delivery, neonatal death, stunting, severe stunting and underweight. It trails the incumbent on three targets and leads on two by 0.0005 and 0.0046 PR-AUC, only 0.5% and 5.6% of the corresponding interval widths. Its severe-stunting ROC-AUC is 0.711, but the PR-AUC ranking is unchanged. The same performance range under a different inductive bias supports an informational, rather than architectural, ceiling.

C. Robustness and Secondary Analyses

The ensemble improved three targets by at most 0.002 and was slightly worse on two. This agrees with Table V: once the data set the error ceiling, model diversity leaves little error to cancel. A tuned nine-model panel previously reached ROC-AUC 0.848, 0.769, 0.694, 0.699 and 0.678 for the five targets; the current untuned panel reached 0.845, 0.773, 0.707, 0.700 and 0.688. The change in algorithm panel does not alter the conclusion.

The person-period formulation expanded 104,025 births into 388,016 rows. The estimated mortality hazard was 3.99% in month one, 1.49% in months 1–12, 0.46% in months 12–24 and 0.65% in months 24–60. A child observed at 30 months is censored after three completed intervals, not treated as a five-year survivor. Multi-task learning improved the no-sharing ablation by 0.010 mean ROC-AUC but remained below single-task CatBoost.

D. Temporal Validation

Models trained on 2017–18 preserve much of their ranking ability on 2022, but their probabilities do not transport well (Table VI).

TABLE VI
TEMPORAL VALIDATION: TRAIN 2017–18, TEST 2022

Target	Int.	Temp.	Δ	Pred./Obs.
facility delivery	0.839	0.796	−0.043	0.88
neonatal death	0.766	0.687	−0.079	5.90
stunted	0.708	0.701	−0.007	1.57
severe stunting	0.686	0.682	−0.004	2.81
underweight	0.677	0.672	−0.005	1.44

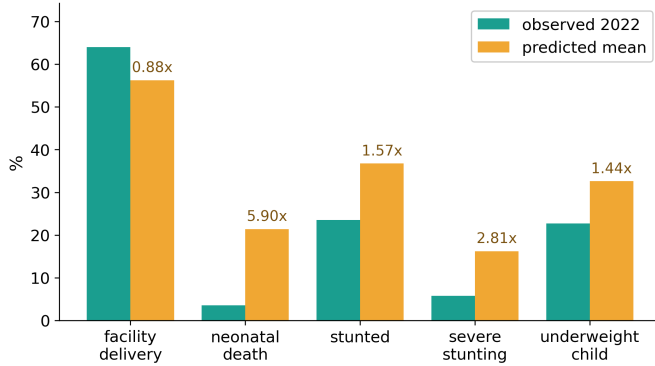


Fig. 3. Calibration drift across survey rounds. Bars compare observed 2022 prevalence with mean risk from a model trained on 2017–18; labels give the ratios in Table VI. The companion internal-versus-temporal ROC-AUC panel was omitted because it repeated columns two and three of the table.

The nutrition targets lose only 0.004–0.007 ROC-AUC across rounds, yet predicted prevalence is 1.44 to 5.90 times observed prevalence. For neonatal death, mean predicted risk is 21.4% versus an observed 3.6%. The model may still rank cases for triage after transport, but probabilities and thresholds need recalibration.

E. Subgroup Performance

Figure 4 compares ROC-AUC and recall at a common 30% flag rate by wealth quintile. Stunting recall is 0.685 in the poorest and 0.121 in the richest, a 5.7-fold difference. For facility delivery, recall is 0.119 in the poorest and 0.812 in the richest. Discrimination is steadier than thresholded sensitivity, so a global cut-point needs stratum-specific reporting or threshold stratification [18].

F. Operational Implications

The 30% flag rate is a capacity scenario, not a clinical recommendation. With no cap, the tenfold false-negative penalty for neonatal death would flag most births, which is impractical. Under the cap, the output is a ranked referral list rather than a diagnosis. A service could recalibrate on a small current-round sample and choose a threshold with its staffing and referral capacity.

VI. DISCUSSION

The main finding is that survey information, not model capacity, limits performance. The seven linear, tree-based and neural learners remain within 0.07 ROC-AUC on every target.

Leakage auditing and maternal-grouped validation explain more of the gap from implausibly high published values than changing the learner.

Limitations are structural: DHS lacks blood pressure, glucose and biomarkers; the records are retrospective; and 65.7% of the measured-child cohort comes from 2017–18. Results are associational and need prospective validation. Future work should add facility supply variables, test recalibration on a held-out period, and assess whether risk-ranked triage improves outcomes.

Nutrition targets are correlated (pairwise phi 0.39–0.51) and are not independent clinical syndromes. The survey does not show whether flagged women received effective interventions, and the transport experiment covers one historical gap rather than future policy or climate shocks. All learners used defaults; broader search might improve a score but would not remove the need for leakage, temporal and subgroup audits.

A. Recommended Reporting Practice

Future DHS prediction studies should report the valid universe, record count and positive rate for each target rather than one pooled denominator. Predictors should be labelled by information time (before contact, same interview, or after outcome) so leakage is auditable. Repeated mothers, siblings and survey clusters should also define resampling groups; a random row split is not independent when households repeat.

Evaluation should include internal and temporal discrimination, calibration or predicted-to-observed prevalence, and thresholded sensitivity by relevant socio-economic strata. Global ROC-AUC alone cannot establish safe transport or equity. Screening reports should state the flag-rate constraint, false-negative cost, recalibration sample and referral capacity with the threshold.

B. Reproducibility and Audit Trail

Data preparation, exploration, modelling and testing are separate rerunnable stages. The cleaned Parquet table, column dictionary, target registry and model manifest are retained. The manifest records each target’s feature block, leakage blacklist, split groups and model family, so a changed valid universe changes the cohort definition rather than every model silently.

Before training results are interpreted, EDA compares weighted prevalence with published round-specific indicators, inspects missingness by module and cohort, and checks correlations for definitional duplicates. Testing stores predictions, bootstrap indices, threshold decisions and subgroup labels, so temporal and wealth analyses reuse the aggregate predictions. These artefacts also permit one target to be rerun after a new survey round is added.

VII. CONCLUSION

Using household survey variables alone, a two-round leakage-audited BDHS pipeline ranks five maternal and child outcomes with test ROC-AUC from 0.688 to 0.845. Linear, tree-based and neural learners, their greedy ensemble, and a

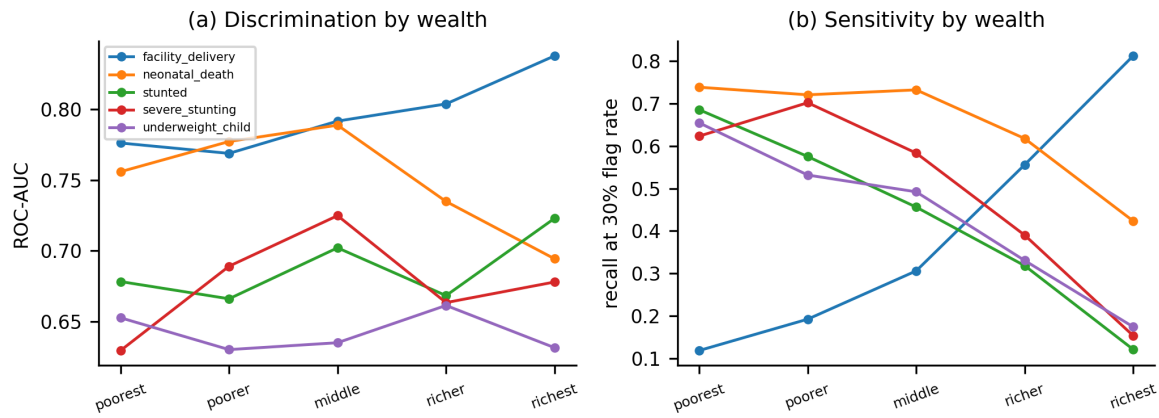


Fig. 4. Subgroup performance by wealth quintile. Discrimination varies less than sensitivity at a single global threshold.

pretrained tabular transformer all fall within this range, placing the ceiling in the data rather than the algorithm.

Two audits determine usability. Temporal transport preserves ranking better than calibration: nutrition targets lose less than 0.007 ROC-AUC, but predicted prevalence is 1.44–5.90 times observed, requiring current-sample recalibration. A global threshold also creates a 5.7-fold sensitivity gap across wealth quintiles, with direction varying by outcome. Thus the practical output is a ranked referral list with a stated flag rate, false-negative cost and stratum-specific sensitivity, not a probability taken at face value.

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REFERENCES

- [1] B. Rao *et al.*, “Machine learning in predicting child malnutrition: a meta-analysis of Demographic and Health Surveys data,” *International Journal of Environmental Research and Public Health*, vol. 22, no. 3, p. 449, 2025.
- [2] M. A. Salam, M. M. Islam, and M. R. Karim, “Machine learning algorithms for predicting and identifying the influencing predictors of antenatal care visits among women in Bangladesh: evidence from BDHS 2022 data,” *PLOS ONE*, 2025.
- [3] S. T. A. Noor, R. B. Islam, S. Yeasar, and S. Siddique, “Machine learning-based prediction of maternal continuum of care completion: evidence from Bangladesh Demographic and Health Survey 2022,” *Array*, 2025.
- [4] S. Zoha *et al.*, “Identifying determinants and predicting cesarean section delivery among Bangladeshi women using machine learning: insight from BDHS 2022 data,” *PLOS Global Public Health*, 2025.
- [5] N. Sultana *et al.*, “Machine learning based prediction of low birth weight and its associated risk factors: insights from the Bangladesh Demographic and Health Survey 2022,” *PLOS Global Public Health*, 2025.
- [6] S. Naznin *et al.*, “Identifying determinants of under-5 mortality in Bangladesh: a machine learning approach with BDHS 2022 data,” *PLOS ONE*, 2025.
- [7] M. M. Islam *et al.*, “Prediction of undernutrition and identification of its influencing predictors among under-five children in Bangladesh using explainable machine learning algorithms,” *PLOS ONE*, vol. 19, no. 12, e0315393, 2024.
- [8] Z. B. Tadese *et al.*, “Prediction of incomplete immunization among under-five children in East Africa from recent demographic and health surveys: a machine learning approach,” *Scientific Reports*, vol. 14, 2024.
- [9] T. Chen and C. Guestrin, “XGBoost: A scalable tree boosting system,” in *Proc. 22nd ACM SIGKDD*, 2016, pp. 785–794.
- [10] G. Ke *et al.*, “LightGBM: A highly efficient gradient boosting decision tree,” in *Advances in Neural Information Processing Systems*, vol. 30, 2017.
- [11] L. Prokhorenkova *et al.*, “CatBoost: unbiased boosting with categorical features,” in *Advances in Neural Information Processing Systems*, vol. 31, 2018.
- [12] N. Erickson *et al.*, “TabArena: a living benchmark for machine learning on tabular data,” in *Advances in Neural Information Processing Systems, Datasets and Benchmarks Track*, 2025.
- [13] R. Caruana, A. Niculescu-Mizil, G. Crew, and A. Ksikes, “Ensemble selection from libraries of models,” in *Proc. 21st Int. Conf. Machine Learning (ICML)*, 2004.
- [14] Y. Gorishniy *et al.*, “RealMLP: Robust MLP for tabular data,” arXiv:2402.06172, 2024.
- [15] Y. Gorishniy *et al.*, “TabM: Advancing tabular deep learning with parameter-efficient ensembling,” in *Proc. ICLR*, 2025.
- [16] N. Hollmann *et al.*, “Accurate predictions on small data with a tabular foundation model,” *Nature*, vol. 637, no. 8045, pp. 319–326, 2025.
- [17] G. S. Collins *et al.*, “TRIPOD+AI statement: updated guidance for reporting clinical prediction models that use regression or machine learning methods,” *BMJ*, vol. 385, e078378, 2024.
- [18] M. Hardt, E. Price, and N. Srebro, “Equality of opportunity in supervised learning,” in *Advances in Neural Information Processing Systems*, vol. 29, 2016.
- [19] National Institute of Population Research and Training (NIPORT), Ministry of Health and Family Welfare, Bangladesh, and ICF, *Bangladesh Demographic and Health Survey 2022*, Dhaka, Bangladesh and Rockville, MD, USA, 2024.